

SF4: Data Logger – Interim Design Report

Smart Guitar Effects Processor & Data Logger

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Group No.: 3

Date: May 26, 2026

1 Introduction & Application Specification

The primary objective of this project is to develop an advanced embedded system utilising the **Arduino Uno** hardware platform (powered by the ATmega328P MCU) configured as an interactive **Smart Guitar Effects Processor and Data Logger**. This directly addresses the requirement to specify an application demonstrating two-way communication between the MCU and PC. A user submits natural-language tone requests (e.g. “Make me sound like David Gilmour”) via a web interface; an LLM translates these into structured DSP parameter JSON payloads that are streamed to the MCU, which returns status/telemetry back to the PC over the same serial link.

2 Conceptual System Design & Communications

The system architecture is strictly partitioned into three specialised operational environments: **Analogue Circuitry**, **Arduino UNO R3 Firmware**, and **Host PC Software**. The block diagram below illustrates the full signal path and bidirectional JSON payload exchange over the 115200 Bd USB Serial Bridge. PC-to-MCU frames carry DSP coefficients (biquad b_0 – a_2 , clip threshold, delay length, PWM duty); MCU-to-PC frames return a compact status packet (clip indicator, current gain, buffer health).

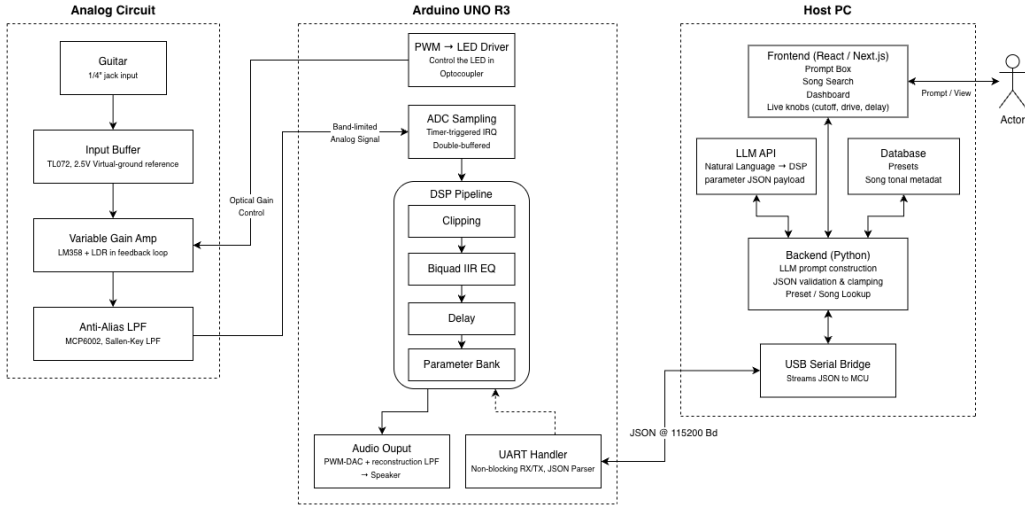


Figure 1: Systems and Communications Block Diagram.

3 Detailed Hardware Design

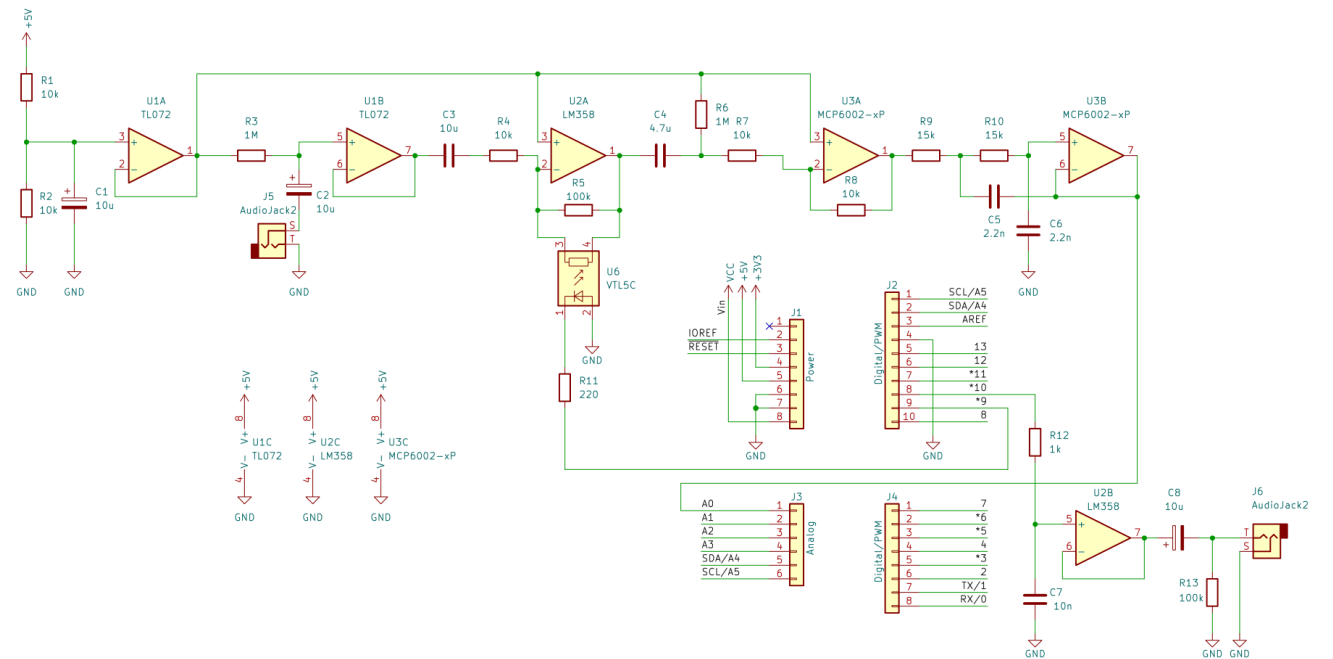


Figure 2: Analogue Circuit Design Schematic (KiCad).

All sub-circuits were unit-tested in LTSpice before integration (see Appendix [6](#)). **Stage 1 (Input Buffer):** TL072 JFET op-amp with 1 M Ω input impedance; second channel buffers a R_1/R_2 divider to produce a 2.5 V virtual-ground reference. **Stage 2 (Variable Gain Amplifier):** LM358 inverting amplifier with an NSL-32SR3

optocoupler LDR in the feedback path; firmware PWM controls the LED to set gain optically, isolating the audio path from digital noise. Simulation confirms variable gain across $R_{LDR} = 100\Omega\text{--}100\text{ k}\Omega$. **Stage 3 (Anti-Aliasing LPF):** MCP6002 Sallen-Key LPF ($R = 15\text{ k}\Omega$, $C = 2.2\text{ nF}$, $f_c \approx 4.8\text{ kHz}$), satisfying the Nyquist criterion for 20 kHz ADC sampling. **Audio Output:** the Arduino PWM DAC output is fed into an RC reconstruction filter (LM358, $R = 1\text{ k}\Omega$, $C = 33\text{ nF}$) which integrates the switching waveform into a smooth analogue sinusoid, then AC-coupled via $10\text{ }\mu\text{F}$ to the speaker load.

4 Firmware & Software Implementation Plan

The firmware architecture circumvents standard blocking Arduino function libraries to sustain low-latency real-time audio operations, utilising `Timer1` interrupts at 20 kHz (50 μs period) for ADC triggering. The Host PC software utilises a React/Next.js frontend and Python/FastAPI backend to handle LLM-driven JSON parameter generation.

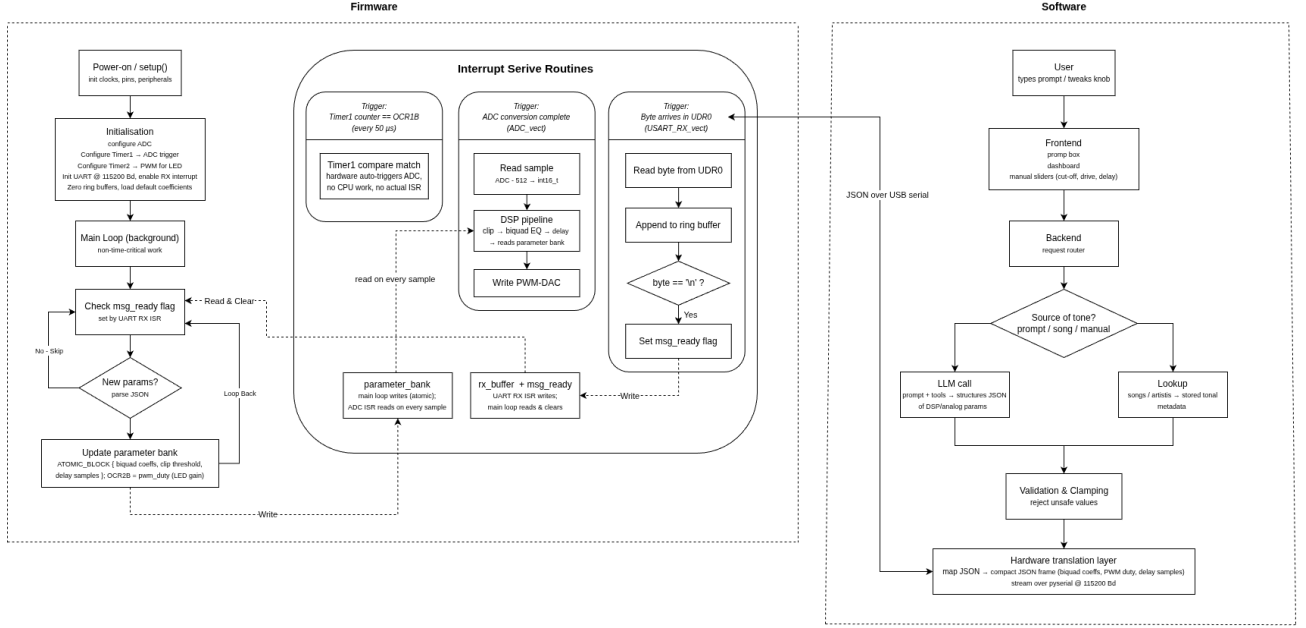


Figure 3: Firmware and Software Flow Diagram.

5 Component Requirements & Parts List

To assemble the target custom prototype circuit board, components have been selected within the strict **£15.00 project budget limit**.

Table 1: Parts List for Purchasing (BOM)

Order Code	Brief Component Description	Qty	Unit Price (£)	Total (£)
SC17220	JFET-Input Dual Op-Amp (TL072CP)	1	1.000	1.000
SC17198	Dual Low-Power Op-Amps (LM358AP)	1	0.370	0.370
1292245	Rail-to-Rail Dual Op-Amp (MCP6002-I/P)	1	0.346	0.346
3692218	Optocoupler / Vactrol (NSL-32SR3)	1	3.290	3.290
CN23320	6.35mm (1/4") PCB Mono Jack Socket	1	0.930	0.930
CN22697	3.5mm Stereo Panel Jack Socket	1	0.330	0.330
Total Custom Expenditure (Budget: £15.00)				£6.266

6 Work Allocation

To optimise execution efficiency across the remaining two-week timeline, work tasks are divided as follows:

- **Gijeong Lee (CRSid: gl534):** Primary ownership of hardware and full-stack software; responsible for breadboarding the analogue front-end, testing the Sallen-Key LPF and VGA stages, calibrating optocoupler gain metrics, and building the Host PC software stack comprising the React/Next.js frontend (prompt UI, dashboard, manual sliders), Python/FastAPI backend (LLM prompt construction, JSON validation), and Supabase database (preset and song tonal metadata storage).
- **Jaehyung Choi (CRSid: jc2409):** Primary ownership of firmware and hardware–software communication; responsible for register-direct `Timer1` ADC triggering, the real-time biquad/clipping/delay DSP pipeline, non-blocking UART interrupt handlers, and the PC-side hardware translation layer (pyserial scripts that serialise validated JSON payloads and parse incoming MCU status packets).

Appendix: LTSpice Unit-Test Schematics and Simulation Results

Each analogue sub-circuit was designed and verified independently in LTSpice before integration into the full circuit (Figure 13). Note: the MCP6002 used in the physical build (Stage 3 LPF) was substituted with the LTC6078 in all LTSpice simulations – the MCP6002 SPICE model repeatedly failed to converge during import. The LTC6078 is a native LTSpice component with near-identical specifications (rail-to-rail output, single-supply, comparable GBW and input offset), making it a valid simulation proxy.

Stage 1 – Input Buffer

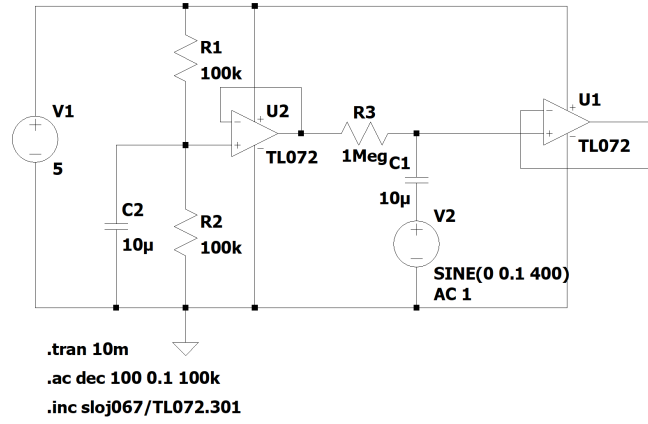


Figure 4: Input buffer LTSpice schematic.

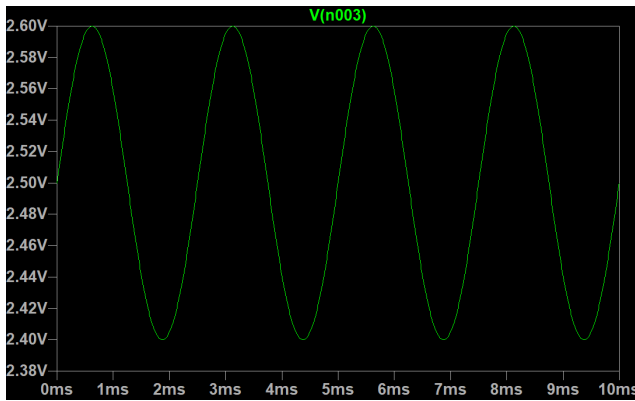


Figure 5: Transient: output biased at 2.5 V, undistorted swing.

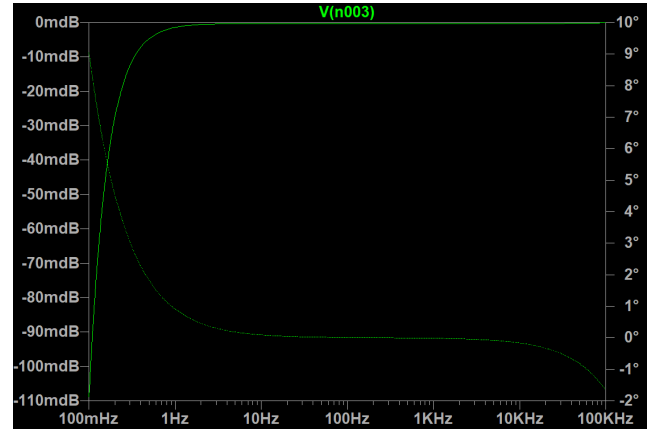


Figure 6: AC sweep: flat 0 dB passband from < 1 Hz to > 100 kHz.

Stage 2 – Variable Gain Amplifier

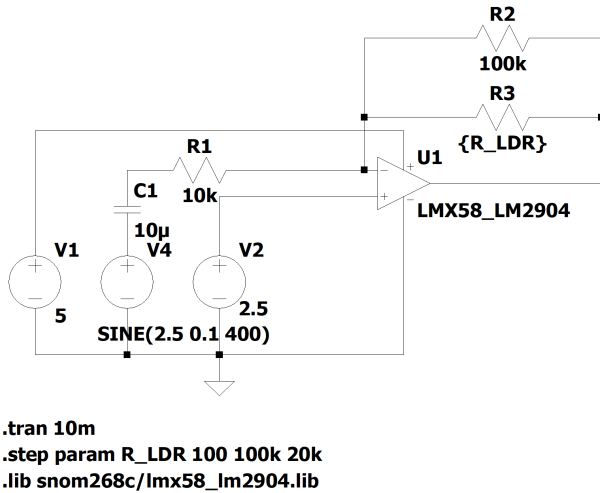


Figure 7: VGA LTSpice schematic (R_{LDR} parametric).

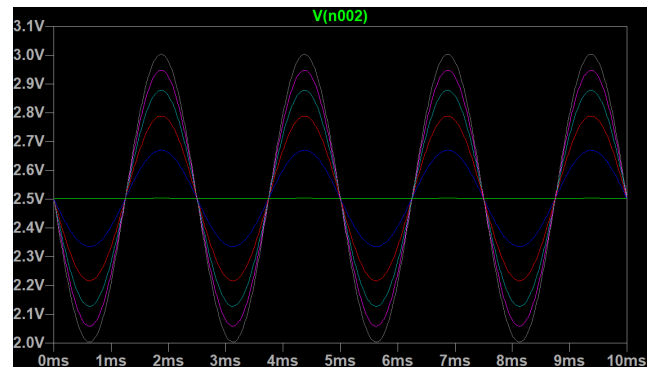


Figure 8: Parametric sweep: gain varies with R_{LDR} (100 Ω –100 k Ω).

Stage 3 – Anti-Aliasing Low-Pass Filter

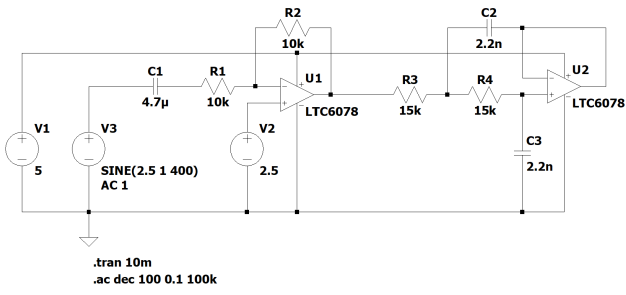


Figure 9: Sallen-Key LPF LTSpice schematic.

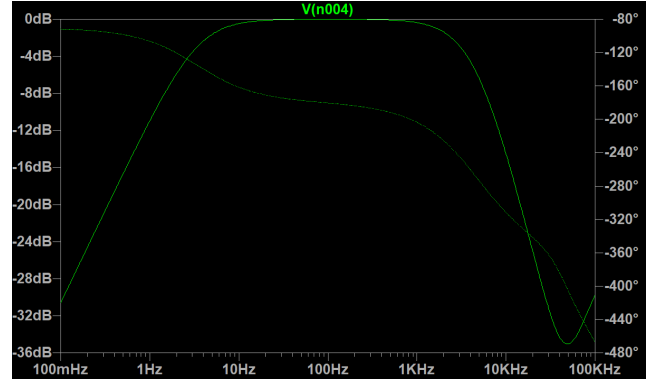


Figure 10: Bode plot: 0 dB passband with rolloff onset at ~ 5 kHz.

Audio Output Stage

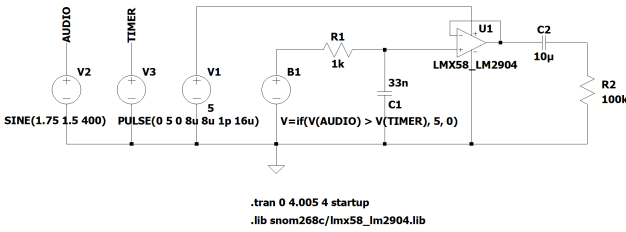


Figure 11: PWM reconstruction filter LTSpice schematic.

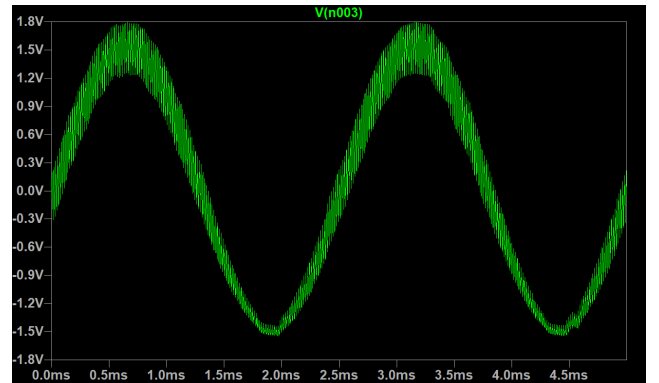


Figure 12: Transient: clean ± 1.5 V sinusoidal output after PWM reconstruction.

Full Circuit Integration Test

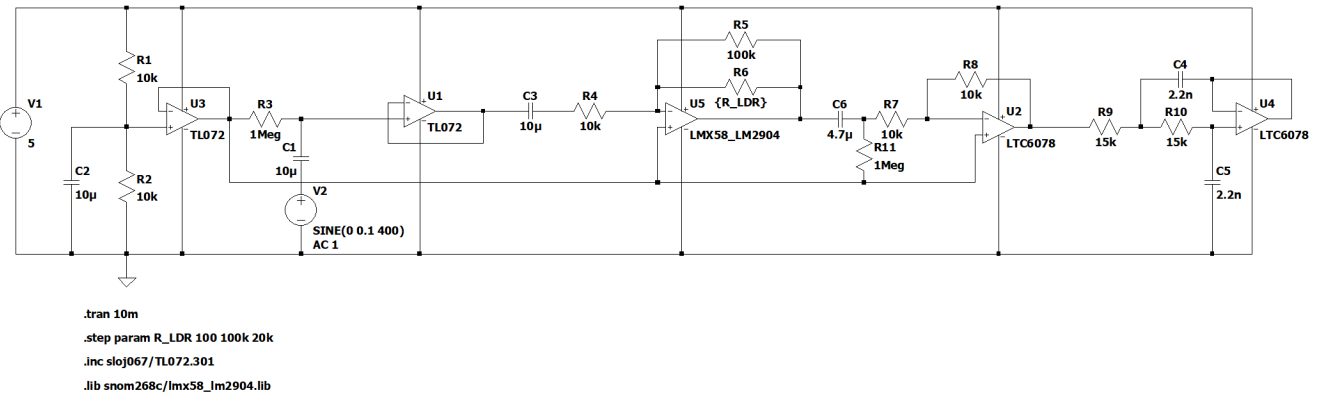


Figure 13: Full LTSpice integration schematic with all stages chained. The output waveform matches the VGA stage, as the 400 Hz test signal lies well within the LPF passband ($f_c \approx 4.8$ kHz) and passes through unattenuated.

Cambridge University Engineering Department
Engineering Tripos Part IIA
PROJECTS: Interim and Final Report Coversheet

IIA Projects

TO BE COMPLETED BY THE STUDENT(S)

Project:	SF4, Data Logger		
Title of report:	SF4: Data Logger – Interim Design Report Smart Guitar Effects Processor & Data Logger		
Name(s): (capitals)	crsID(s):	College(s):	
GIJEONG LEE	gl534	Clare College	
JAEHYUNG CHOI	jc2409	Hughes Hall	
<u>Declaration</u> for: Interim Report			
I/we confirm that, except where indicated, the work contained in this report is my/our own original work.			

Instructions to markers of Part IIA project reports:

Grading scheme

Grade	A*	A	B	C	D	E
Standard	Excellent	Very Good	Good	Acceptable	Minimum acceptable for Honours	Below Honours

Grade the reports by ticking the appropriate guideline assessment box below, and provide feedback against as many of the criteria as are applicable (or add your own). Feedback is particularly important for work graded C-E. Students should be aware that different projects and reports will require different characteristics.

Penalties for lateness: Interim Reports: 3 marks per weekday; Final Reports: 0 marks awarded – late reports not accepted.

Guideline assessment (tick one box)

A*/A	A/B	B/C	C/D	D/E

Marker:		Date:	
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Delete (1) or (2) as appropriate (for marking in hard copy – different arrangements apply for feedback on Moodle):

- (1) Feedback from the marker is provided on the report itself.
- (2) Feedback from the marker is provided on second page of cover sheet.

	Typical Criteria	Feedback comments
Project Skills, Initiative, Originality	Appreciation of problem, and development of ideas	
	Competence in planning and record-keeping	
	Practical skill, theoretical work, programming	
	Evidence of originality, innovation, wider reading (with full referencing), or additional research	
	Initiative, and level of supervision required	
Report	Overall planning and layout, within set page limit	
	Clarity of introductory overview and conclusions	
	Logical account of work, clarity in discussion of main issues	
	Technical understanding, competence and accuracy	
	Quality of language, readability, full referencing of papers and other sources	
	Clarity of figures, graphs and tables, with captions and full referencing in text	